
Shunting Inhibition Improves Local Credit Assignment in Dendritic Networks

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Abstract

Exact gradients for conductance-based dendritic trees factorize into a synapse-local eligibility term and a single non-local compartment error; this factorization is the foundation for local credit assignment in these networks. How well a somatic broadcast can substitute for the true compartment error depends on how concentrated the path gains are across the tree. An oracle experiment—replacing the broadcast with the exact transported compartment errors—raises additive local learning from 33% to 98.5% on noise resilience, nearly matching shunting (98.7%), strongly supporting path-gain fidelity as the dominant bottleneck rather than other architectural differences. Simple per-soma broadcast already improves gradient fidelity substantially: cosine alignment rises from 0.006 to 0.202 and noise_resilience accuracy from 0.33 to 0.90 under shunting relative to additive integration. The advantage is regime-dependent, growing with inhibitory conductance and largest under noisy inputs. A structured pathway-vector extension (PV-LocalCA) further stabilizes learned-routing performance on a hard cue-integration benchmark, reaching 99.1% test accuracy. Together, these results identify path-gain concentration via shunting inhibition as the key biophysical mechanism that determines when low-bandwidth local credit assignment works in conductance-based dendritic networks.

1 Introduction

Credit assignment in deep networks usually depends on backpropagation: global error transport through symmetric pathways with no obvious biological substrate. Dendritic compartments suggest an alternative. Each synapse has access to rich local state such as driving force, membrane conductance, and branch voltage, while supervisory information could arrive as a low-bandwidth broadcast from the soma [15]. The question in this paper is whether such local information is sufficient for useful learning.

We show that it can be, but only in a specific biophysical regime. Starting from conductance-based dendritic voltage equations [1], we derive exact gradients for dendritic trees (Theorem 1) and show that each synaptic gradient factorizes into purely local terms and a single non-local compartment error. Approximating that compartment error with a broadcast signal yields a family of local learning rules (3F, 4F, and 5F) that use only branch-local variables together with the broadcast. For tasks where different branches must be credited differently, we further introduce a structured pathway-vector signal that preserves latent pathway identity during credit assignment.

The central mechanistic finding is that shunting improves broadcast fidelity by concentrating path gains. Path-gain CV stays at 0.21–0.27 under shunting versus 0.53–1.08 under additive integration on MNIST, meaning that one shared broadcast is a far better proxy for compartment-specific errors

under shunting. Correspondingly, per-soma broadcast achieves cosine alignment of 0.20–0.32 under shunting on the noise-resilience sweep, compared to near-zero or negative alignment under additive integration. Our question is not whether local learning can match backpropagation on standard benchmarks, but what biophysical mechanism determines when low-bandwidth local credit assignment works in conductance-based dendritic networks. An oracle experiment that supplies the exact transported compartment errors strongly supports path-gain fidelity as the dominant bottleneck: additive local learning rises to 98.5% on noise resilience under the oracle, nearly matching shunting (98.7%). The advantage is regime-dependent, growing with inhibitory conductance and largest under noisy inputs.

We quantify these effects using a gradient-fidelity diagnostic based on directional alignment (cosine similarity) and scale mismatch between local and backprop gradients (Table 1, Fig. 2). The absolute cosine values remain modest, but they are consistently more positive under shunting and predict which local-learning regimes ultimately succeed. The shunting advantage is regime-dependent: it grows with inhibitory conductance and is largest on tasks where credit signals must remain robust under noise (Fig. 4). On standard classification benchmarks, simple per-soma broadcast is often sufficient. On the hard cue-routing benchmark, however, scalar broadcast collapses distinctions between competing branches, motivating the structured pathway-vector rule studied in Sec. 4.7.

Contributions

1. **Exact gradients for conductance-based dendritic trees.** We derive exact loss gradients for dendritic trees and show that each synaptic gradient factorizes into a synapse-local eligibility term and a compartment-specific error.
2. **A family of local learning rules.** We use this factorization to define 3F/4F/5F local updates based on synapse-local variables and a somatic broadcast, and extend this family with PV-LocalCA when pathway identity must be preserved.
3. **A regime-dependent advantage for shunting.** In matched architectures, shunting improves gradient fidelity and local learning relative to an additive control, with the largest gains in noisy or inhibition-dependent settings.
4. **A gradient-fidelity diagnostic.** We use a component-wise local-vs-backprop diagnostic to link dendritic architecture and broadcast design to the quality of local credit signals.

2 Compartmental Voltage Model and Gradient Derivation

We use a steady-state conductance model derived from discretized passive cable dynamics [1, 2]. In normalized units, leak and inhibitory reversal are set to 0, the excitatory reversal is set to 1, and leak conductance is fixed at 1. Each compartment voltage is therefore a conductance-weighted average. Two properties of this form matter for what follows: local sensitivities depend on both driving force ($E - V$) and input resistance R^{tot} , and shunting inhibition corresponds to adding conductance when $E_{\text{inh}} \approx 0$.

2.1 Voltage Equation and Local Sensitivities

Consider compartment n with synaptic inputs j (activity x_j , reversal E_j , conductance $g_j^{\text{syn}} \geq 0$) and dendritic inputs from children (voltage V_j , conductance $g_j^{\text{den}} \geq 0$). The steady-state voltage is:

$$V_n = \frac{\sum_j E_j x_j g_j^{\text{syn}} + \sum_j V_j g_j^{\text{den}}}{\underbrace{\sum_j x_j g_j^{\text{syn}} + \sum_j g_j^{\text{den}} + 1}_{g_n^{\text{tot}}}}, \quad R_n^{\text{tot}} = 1/g_n^{\text{tot}}. \quad (1)$$

V_n is a convex combination of reversal potentials, child voltages, and leak. Writing $\mathcal{S}_n = \{0\} \cup \{E_j\}_j \cup \{V_j\}_j$, we have $\min \mathcal{S}_n \leq V_n \leq \max \mathcal{S}_n$ and $0 < R_n^{\text{tot}} \leq 1$. The local sensitivities follow directly:

Proposition 1 (Local Sensitivities).

$$\frac{\partial V_n}{\partial g_i^{\text{syn}}} = x_i R_n^{\text{tot}} (E_i - V_n), \quad \frac{\partial V_n}{\partial V_i} = g_i^{\text{den}} R_n^{\text{tot}}, \quad \frac{\partial V_n}{\partial g_i^{\text{den}}} = R_n^{\text{tot}} (V_i - V_n). \quad (2)$$

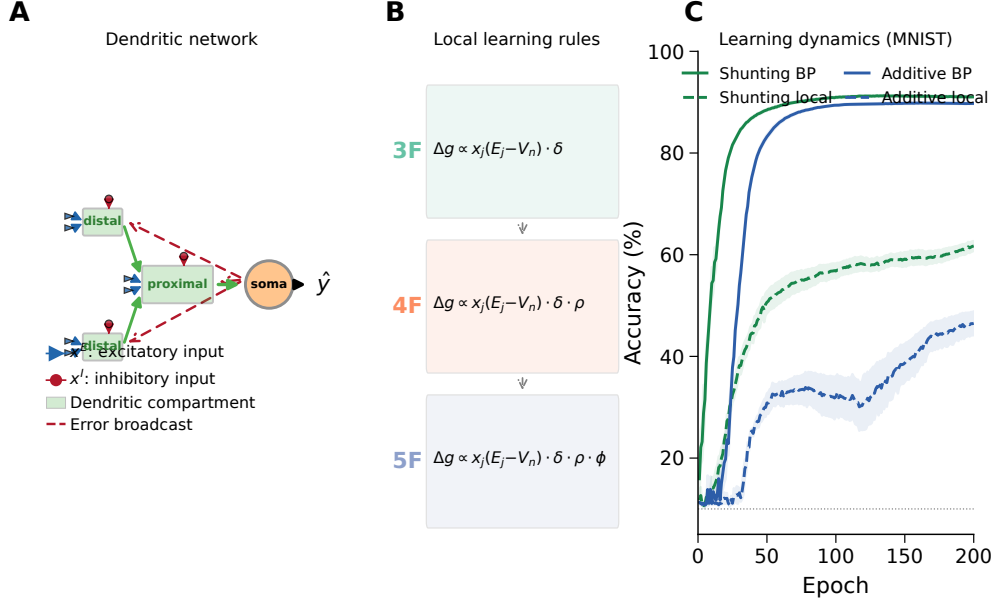


Figure 1: Model and credit assignment. (A) A compartmental dendritic neuron with excitatory (blue triangles) and inhibitory (red circles; shunting) synapses. Voltages propagate to the soma through learned dendritic conductances (green arrows). A somatic broadcast error e_n (dashed red) reaches all compartments; each synapse combines e_n with local quantities (presynaptic drive x_j , input resistance R_n^{tot} , and driving force $E_j - V_n$). (B) Compact LocalCA hierarchy: 3F uses local eligibility and a broadcast field; 4F adds the morphology-aware factor ρ ; 5F further adds the confidence-like factor ϕ . (C) Broadcast-mode schematic. Scalar broadcast sends one shared field to all branches; per-soma broadcast sends a vector-valued somatic error to each branch; structured pathway feedback uses separate channels to preserve pathway identity when different branches should receive different teaching signals. (D) Representative fixed-100-epoch MNIST learning curves for shunting BP, 3F, 4F, and 5F. All three local variants learn substantially above chance, while BP remains the strongest reference. In this representative run, 4F slightly exceeds 5F, underscoring that Fig. 1 is meant to illustrate the rule-family progression rather than replace the sweep-aggregated ranking in Appendix Table S1.

2.2 Shunting Inhibition as Divisive Gain Control

An inhibitory synapse with $E_{\text{inh}} \approx 0$ contributes current $(0 - V_n)x_j g_j^{\text{syn}}$ and increases g_n^{tot} . Its sensitivity is $\partial V_n / \partial g_j^{\text{syn}} = -x_j R_n^{\text{tot}} V_n$, which corresponds to multiplicative attenuation (divisive normalization). Shunting is divisive at the voltage level, but its effect on firing rates can be subtractive in some regimes [7]. Inhibitory plasticity can balance excitation dynamically [8]; our learned inhibitory conductances serve an analogous role.

2.3 Exact Gradients for Dendritic Trees

Let V_0 be the somatic output, $\hat{y} = W_{\text{dec}} V_0$ the decoder, and $\delta_0 = W_{\text{dec}}^T (\partial L / \partial \hat{y})$ the somatic error.

Theorem 1 (Backpropagation on a Dendritic Tree). *For a rooted dendritic tree with soma at node 0 and unique parent $p(n)$ for each non-somatic compartment, the loss gradient satisfies*

$$\frac{\partial L}{\partial V_n} = \frac{\partial L}{\partial V_{p(n)}} R_{p(n)}^{\text{tot}} g_{n \rightarrow p(n)}^{\text{den}}, \quad (3)$$

with boundary condition $\partial L / \partial V_0 = \delta_0$. Because the graph is a tree, each compartment has a unique path to the soma. Defining the path gain

$$\alpha_n = \prod_{(i \rightarrow k) \in \text{path}(n \rightsquigarrow 0)} R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}, \quad \frac{\partial L}{\partial V_n} = \alpha_n \delta_0, \quad (4)$$

89 with $\alpha_0 = 1$.

90 *Proof.* Apply the chain rule on the tree-structured computation graph using Prop. 1. $\square$

91 **Corollary 1** (Local–Global Factorization). *The exact synaptic gradient at compartment n factorizes*
 92 *as:*

$$\frac{\partial L}{\partial g_i^{\text{syn}}} = \underbrace{x_i R_n^{\text{tot}} (E_i - V_n)}_{\text{synapse-local eligibility}} \cdot \underbrace{\frac{\partial L}{\partial V_n}}_{\text{compartment error}}, \quad (5)$$

93 *where the eligibility term depends only on quantities available at synapse i (presynaptic activity*
 94 *x_i , input resistance R_n^{tot} , driving force $E_i - V_n$), and the compartment error $\partial L/\partial V_n$ is the sole*
 95 *non-local quantity.*

96 This factorization isolates the only non-local quantity: $\partial L/\partial V_n$. Any practical local rule therefore
 97 depends on how well a broadcast signal approximates that compartment error, which we evaluate
 98 in Sec. 4.2. In the models studied here, shunting improves this approximation by reducing and
 99 stabilizing R_n^{tot} , which narrows the spread of local sensitivities across the tree.

100 3 Local Learning Rules

101 3.1 Broadcast Error Approximation

102 We approximate the exact compartment error $\partial L/\partial V_n$ (Corollary 1) with a broadcast signal e_n
 103 derived from the somatic error $\delta_0 = W_{\text{dec}}^\top (\partial L/\partial \hat{y})$. We consider four broadcast modes of increasing
 104 locality:

- 105 • **Scalar:** $e_n = \bar{\delta} \cdot \mathbf{1}$, where $\bar{\delta} = \text{mean}(\delta_0)$ is a single scalar broadcast to all compartments.
- 106 • **Per-soma:** $e_n = \delta_0$ when the layer dimension matches the output; layers with mismatched
 107 dimensions fall back to scalar.
- 108 • **Local mismatch:** $e_n = (1-\alpha) \bar{\delta} \cdot \tilde{m}_n + \alpha \bar{\delta}$, where $\tilde{m}_n$ is the RMS-normalized, batch-
 109 centered parent–child voltage difference $P_n - V_n$ and $\alpha = 0.2$.
- 110 • **Pathway-vector:** $e_n = \alpha \bar{\delta} \cdot \mathbf{1} + (1-\alpha) \Gamma_n(\delta_0)$, where Γ_n projects the somatic error into
 111 router-inferred pathway roles, gates those roles by local pathway activity, and transports the
 112 resulting vector hierarchically through the block-structured dendritic tree.

113 We use per-soma broadcast by default. In our experiments, the local-mismatch variant performs
 114 much worse (Appendix A), indicating that broadcast quality is critical. Under shunting, this simple
 115 broadcast is often sufficient on standard classification tasks, but the cue-routing benchmark in Sec. 4.7
 116 requires the structured pathway-vector form. Appendix Table S6 summarizes the broader set of
 117 implemented model families, training strategies, and broadcast modes used throughout the study.

118 3.2 Three-Factor Rule (3F)

119 **Definition 1** (3F Update). *For synaptic and dendritic conductances:*

$$\Delta g_j^{\text{syn}} = \eta \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B, \quad \Delta g_j^{\text{den}} = \eta \langle R_n^{\text{tot}} (V_j - V_n) e_n \rangle_B, \quad (6)$$

120 *where $\langle \cdot \rangle_B$ denotes the batch average.*

121 The three factors are presynaptic activity x_j (or a voltage difference), postsynaptic modulation
 122 through driving force and input resistance, and the broadcast error e_n . The same rule applies to
 123 excitatory and inhibitory synapses. The sign difference comes from the driving force $(E_j - V_n)$.

124 **Additive control.** Rule (6) is the local gradient of the *shunting* voltage $V_n = \sum E_j g_j x_j / g_n^{\text{tot}}$. For
 125 the additive control ($V_n = \sum g_j x_j$, no divisive normalization), the correct local gradient is simpler:
 126 $\Delta g_j^{\text{syn}} = \eta \langle x_j e_n \rangle_B$, with no driving-force or R_n^{tot} terms ($R_n^{\text{tot}} \equiv 1$ by definition since there is no
 127 denominator). Throughout, each architecture uses the learning rule derived from its own forward-pass
 128 dynamics. This keeps the comparison architecture-matched rather than applying a shunting-derived
 129 rule to an additive model.

3.3 Higher-Order Rules: 4F and 5F

4F (heuristic attenuation proxy). In the exact gradient (4), the path gain α_n attenuates the somatic error differently at each compartment. To compensate without explicitly computing that gain, we use an online correlation proxy

$$\rho_n = \text{Cov}(\bar{V}_n, \bar{V}_0) / (\sqrt{\text{Var}(\bar{V}_n) \text{Var}(\bar{V}_0)} + \varepsilon).$$

High ρ_n indicates that compartment and somatic activity covary strongly, making the broadcast δ_0 a more plausible proxy for the local compartment error; low ρ_n down-weights updates where that proxy is likely to be poor. ρ_n is estimated online via exponential moving average ($\alpha = 0.1$).

5F (confidence-weighted 4F). The current 5F implementation adds a second heuristic factor

$$\phi_n = \text{Var}(V_n) / (\sigma_{\text{res}}^2 + \varepsilon),$$

where σ_{res}^2 is the residual variance of V_n after linear regression on the parent voltage. This is an inverse residual-fraction or signal-to-noise proxy: it becomes large when compartment activity is strong and stable relative to residual fluctuations unexplained by the parent, and small when local fluctuations dominate. Operationally, ϕ_n behaves as a branch-wise confidence gate rather than as a pure unique-information estimator. The large empirical gap between 4F and 5F in our sweeps indicates that this confidence reweighting removes a major source of update noise: 3F/4F treat all compartments with similar broadcast quality similarly, whereas 5F suppresses branches whose local voltage is dominated by residual variability. When σ_{res}^2 is very small, ϕ_n would otherwise become degenerate, so we clamp it to a finite interval and interpret saturated values simply as “high-confidence” branches rather than as precise estimates of information content. Clamped to $[0.25, 4.0]$ for stability.

It is important to note that ϕ_n is an empirical heuristic modulator, not a derived consequence of the conductance-tree gradient factorization in Theorem 1. The paper’s core mechanistic claim rests on the path-gain concentration result and the oracle experiment; the performance improvement from ϕ_n should be interpreted as engineering value rather than theoretical necessity.

Proposition 2 (5F Update).

$$\Delta g_j^{\text{syn}} = \eta \rho_n \phi_n \langle x_j R_n^{\text{tot}} (E_j - V_n) e_n \rangle_B. \quad (7)$$

Gradient alignment under random broadcast. When the broadcast matrix B_n has i.i.d. zero-mean entries with $\mathbb{E}[B_n^\top B_n] = \alpha I$, the expected cosine between local and exact gradients is positive: $\mathbb{E}[\cos \angle(g^{\text{local}}, g^{\text{exact}})] \geq c_n > 0$, by an argument analogous to feedback alignment [9]. The constant c_n depends on the correlation between local factors and the exact path gain (4); shunting architecture increases this correlation by normalizing the scale of intermediate signals.

3.4 Structured pathway feedback and branch-role plasticity

For tasks with latent pathway structure, scalar broadcast can erase the distinction between different dendritic branches that should be credited differently. Recent experimental evidence supports the existence of vectorized instructive signals in cortical dendrites [47], motivating the structured feedback variant we introduce here. We therefore augment the local rule with a structured feedback variant used only on the cue-routing benchmark. We refer to this structured cue-routing rule as *PV-LocalCA* (pathway-vector LocalCA).

Hierarchical pathway-vector broadcast. The encoder/router defines a low-dimensional role basis over latent pathways (for example, one role per cue stream together with shared context). Instead of collapsing δ_0 to a scalar or per-soma vector, we project it into this role basis, gate each role by the local pathway activity on the target branch, and propagate the resulting vector through the proximal-to-distal block tree. Earlier dendritic layers therefore receive a pathway-specific teaching signal that remains aligned with their downstream descendants. The role coefficients $c_k = \langle r_k, \delta_0 \rangle$ project the somatic error into the k -th router-inferred pathway role r_k , and the local branch weights $q_{n,k}$ are given by the softmax-normalized router assignment of branch n to pathway k , estimated from incoming block weights. Writing the low-rank role coefficients as c_k and the local branch role weights as $q_{n,k}$, the structured broadcast takes the form

$$e_n^{\text{PV}} = \sum_{k=1}^K q_{n,k} c_k,$$

with K latent pathways. Scalar broadcast is the special case $K=1$, while per-soma broadcast corresponds to a dense output-aligned basis.

Router-derived branch roles. Rather than assigning hand-coded apical/basal labels, each branch j receives a normalized role profile r_j inferred from router assignments and the branch’s incoming block weights. Plasticity then depends on two additional local quantities: a branch selectivity factor that up-weights branches with concentrated role mass, and a synapse-specific alignment factor proportional to $\langle r_i, r_j \rangle$ for pre/post role profiles. This biases learning toward pathway-consistent afferents without imposing a heuristic branch taxonomy.

4 Experiments

4.1 Setup

We evaluate the model in two experimental settings: a *capacity-calibrated* regime, where the same architectures achieve strong performance under backpropagation, and a set of *controlled sweeps* that isolate the effects of inhibition strength, broadcast mode, and architecture on credit-signal quality. Our main datasets are MNIST, Fashion-MNIST [40], and four synthetic tasks: *context gating* (context-dependent category boundaries), *noise resilience* (learning under structured input noise), *info shunting* (a task requiring inhibition-mediated gating), and *cue integration* (two competing sensory cues plus context indicating which cue is reliable); formal task definitions are in Appendix I. The cue-integration benchmark is specifically designed to test whether dendritic pathway structure can support context-dependent reliability weighting rather than simple averaging: a fixed-pathway control duplicates context into each cue pathway, while a learned router must discover that separation from data. Architectures include point MLP baselines and dendritic cores with either additive integration or shunting (conductance-based) inhibition. Our primary credit-assignment comparisons are architecture-matched additive vs. shunting cores and local vs. backprop training; DFA is included as a supplementary feedback baseline where the layer geometry permits it. Most headline MNIST, Fashion-MNIST, and context-gating results use the 5F rule with per-soma broadcast. The hard cue-routing benchmark instead uses PV-LocalCA: a structured 3F variant with pathway-vector broadcast, path propagation, inhibitory homeostasis, and router-derived branch-role plasticity. For the inhibition sweeps, we additionally evaluate a transported-error oracle that recursively applies the exact path gains from Theorem 1; this oracle is used only as an upper-bound control, not as the main biological proposal. Because context gating benefits from a small HSIC auxiliary objective (about 3 percentage points in Appendix Fig. S4C), we treat it as a supporting structured benchmark rather than as the cleanest test of the base LocalCA rule. We report means $\pm$ s.d. across 3–5 seeds for all headline results. We will release code and configuration files after the review process. The experiments are organized around one mechanistic chain: we first verify how closely different broadcasts track the exact gradients implied by Theorem 1, then test whether those fidelity differences translate into learning, and finally study a structured-feedback extension for settings where scalar or per-soma broadcast fails.

4.2 From Exact Factorization to Credit-Signal Fidelity

To test whether these performance differences correspond to better credit signals, we compare local and backprop gradients on the same batch at the same parameter values. For each parameter tensor p , we compute directional alignment (cosine similarity) and scale mismatch, defined as $|\log_{10}(\|g_p^{\text{local}}\|/\|g_p^{\text{bp}}\|)|$, aggregated by parameter count.

Alignment dynamics over training. Figure 2B tracks per-layer cosine similarity over epochs. In shunting networks, proximal layers approach high positive alignment over training, whereas distal layers show smaller but still positive alignment. Additive networks show near-zero or negative alignment at all layers and epochs. Component-wise decomposition (Fig. 2C) reveals that dendritic conductances and excitatory synapses carry the strongest alignment signal in shunting networks, consistent with the biophysical role of conductance-based driving forces. Although the absolute cosines are still far from 1, they are strongly predictive of local-learning competence: the same inhibition regimes that raise these cosines are the regimes in which local training succeeds.

Dataset	Core	Weighted cosine $\uparrow$	Scale mismatch $\downarrow$
MNIST	Shunting	0.202	0.117
MNIST	Additive	0.006	1.053
Context gating	Shunting	0.108	0.036
Context gating	Additive	-0.007	2.154

Table 1: **Gradient fidelity (5F + per-soma).** Local vs. backprop gradients on matched weights. Shunting shows substantially higher directional alignment and lower scale mismatch than the additive control; the relative gains are large even though the absolute cosines remain modest.

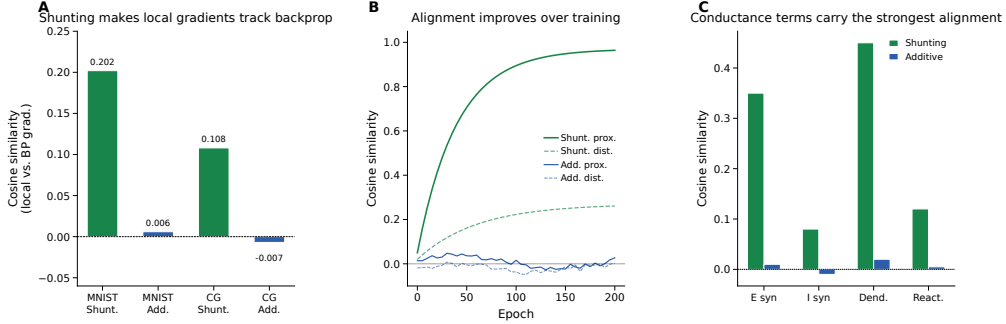


Figure 2: **Gradient-fidelity mechanism.** (A) Weighted cosine similarity between local and backprop gradients: shunting (green) is consistently more aligned than additive (blue) on both MNIST and context gating. (B) Per-layer cosine similarity over training epochs. Shunting proximal layers approach high positive alignment; additive layers remain near zero. (C) Component-wise alignment: E-synapses and dendritic conductances carry the strongest alignment signal in shunting networks.

224 **Exact factorization sanity check.** Using the exact compartment errors $\partial L/\partial V_n$ extracted by auto-
225 grad, we reconstructed the corresponding synaptic and dendritic conductance gradients directly from
226 the factorization in Corollary 1, using the exact piecewise denominator derivative of the implemented
227 shunting layer. Across the full 100-run inhibition sweep used for the mechanistic analysis, the recon-
228 structed gradients match autograd to numerical precision (weighted cosine 1.000001 ± 0.000001 ,
229 weighted scale mismatch $< 4 \times 10^{-8}$). This confirms that the theory is operational in code: the
230 only approximation made by LocalCA is in the broadcast substitute for $\partial L/\partial V_n$, not in the local
231 eligibility factors.

232 4.3 Mechanistic Chain: Path Gains, Broadcast Fidelity, and Oracle Transport

233 Figure 3 provides the paper’s central empirical chain: shunting concentrates path gains, which
234 improves direct compartment-error fidelity, which in turn makes low-bandwidth local learning
235 effective; the transported-error oracle then isolates how much of the remaining gap is due to broadcast
236 quality rather than other architectural differences.

237 **Path-gain concentration.** For a tree, the exact compartment error is the somatic error multiplied
238 by a unique path gain $\alpha_n = \prod_{k \in \text{path}(n \rightarrow 0)} R_k^{\text{tot}} g_k^{\text{den}}$. Panel A measures the coefficient of variation
239 of these path gains across compartments. On MNIST, shunting keeps the path-gain distribution tightly
240 concentrated across inhibition levels (CV about 0.21–0.27), whereas the additive control shows much
241 broader dispersion (CV about 0.53–1.08). This is the deterministic quantity that scalar or per-soma
242 broadcast is implicitly trying to ignore: when the omitted path gains vary less, one shared broadcast
243 is a better proxy for the true compartment-specific error.

244 **Direct compartment-error fidelity.** The theorem identifies $\partial L/\partial V_n$ as the sole non-local term, so
245 we measure it directly. Panel B compares the local broadcast e_n with the exact compartment error
246 on the noise-resilience sweep, using cosine similarity at the compartment level. Per-soma broadcast
247 tracks the exact compartment error much better under shunting than under additive integration, rising
248 from 0.20 at $N_I=0$ to about 0.32 near the best-performing inhibition range, while additive stays near

Dataset	BP ceiling	Best local (5F)	Gap
MNIST	0.965	0.914 ± 0.003	5.1%
Fashion-MNIST	0.879	0.811 ± 0.012	6.8%
Context gating	0.864	0.803 ± 0.006	6.1%

Table 2: **Local competence.** Backprop ceilings from capacity sweeps; local values are 5F with per-soma broadcast on shunting dendritic cores. Context gating additionally uses HSIC auxiliary objective (weight 0.01; Appendix E). Errors: ± 1 s.d. across 5 seeds.

zero or negative. The transported-error oracle obtained by recursively applying the exact path gains improves fidelity further in both architectures, showing that the theorem is not merely descriptive: it defines an operational upper bound on low-bandwidth broadcast. The large gap between the additive per-soma broadcast and its transported upper bound shows that omitted path gains are a major part of the additive failure mode.

Transported broadcast is an operational upper bound. Panel C trains the same local rule using the transported-error oracle as its teaching signal. On noise resilience, additive learning rises from 33–47% under per-soma broadcast to 75.6% at $N_I=0$ and 98.5% at $N_I=40$ under transported broadcast. Shunting improves from 71.7–90.3% under per-soma to 98.5–99.3% under transported broadcast across the same sweep. On MNIST, transported broadcast reaches 94.6% for additive and 96.8% for shunting. These oracle results strongly support path-gain fidelity as the dominant bottleneck for LocalCA performance. The remaining additive–shunting gap under matched transported broadcast (94.6% vs. 96.8% on MNIST) is smaller but still present, suggesting conductance-shaped local sensitivities contribute an additional benefit beyond broadcast fidelity alone.

Low-bandwidth broadcast suffices. Testing degraded broadcast modes (Fig. 3D), 4-bit quantization preserves full performance (61.6% vs. 61.7%; $p > 0.9$), while sign-only (37.5%) and sparse top-30% (38.2%) degrade sharply. Here “effective bits per neuron” means the precision of the broadcast channel after compression: 32 for full-precision, b for b -bit quantization, 1 for sign-only, and a separate sparse code for the top-30% condition. These results suggest that broadcast direction matters more than exact magnitude once local sensitivities are well scaled.

Normalization-only controls remain insufficient. Explicitly normalizing additive voltages improves local learning but does not close the gap to shunting (Appendix Fig. S7). DFA baselines (Appendix Fig. S5) likewise improve over point-MLP random-feedback baselines without matching the shunting conductance model. These controls support the same conclusion as panel B: useful local learning requires more than a broadcast signal with the right overall scale.

4.4 Capacity-Calibrated Competence

Having established the mechanism, we next ask whether the same conditions improve end-task learning in architectures that are already competent under backpropagation. In the capacity-calibrated regime, shunting dendritic cores reach 0.965 on MNIST and 0.864 on context gating under backpropagation. Within the same architectures, the best local configuration (5F, per-soma broadcast, local decoder) reaches the accuracies reported in Table 2.

Within the local-rule family, 5F consistently outperforms 4F and 3F (Appendix Table S1), and per-soma broadcast strongly outperforms scalar and local-mismatch modes (Appendix Table S3). The large 4F→5F jump is important: in our current implementation, most of that gain comes from confidence-weighting updates by ϕ_n , which suppresses noisy branch fluctuations that would otherwise receive the same broadcast-driven update as reliable branches. We therefore interpret 5F as the strongest practical rule in this paper, but not as a uniquely derived optimum; its clamp bounds and residual estimator are still heuristic design choices. In a focused MNIST sensitivity sweep, however, those heuristics are not especially fragile: tightening the clamp from $[0.25, 4.0]$ to $[0.5, 2.0]$ changes mean test accuracy by only +0.4 pp, widening it to $[0.1, 8.0]$ is effectively neutral, and varying the 4F/5F EMA rate from 0.05 to 0.20 shifts accuracy by less than 0.2 pp (Appendix Fig. S3). Appendix Table S2 summarizes the empirical division of labor across architecture, rule, and broadcast choices.

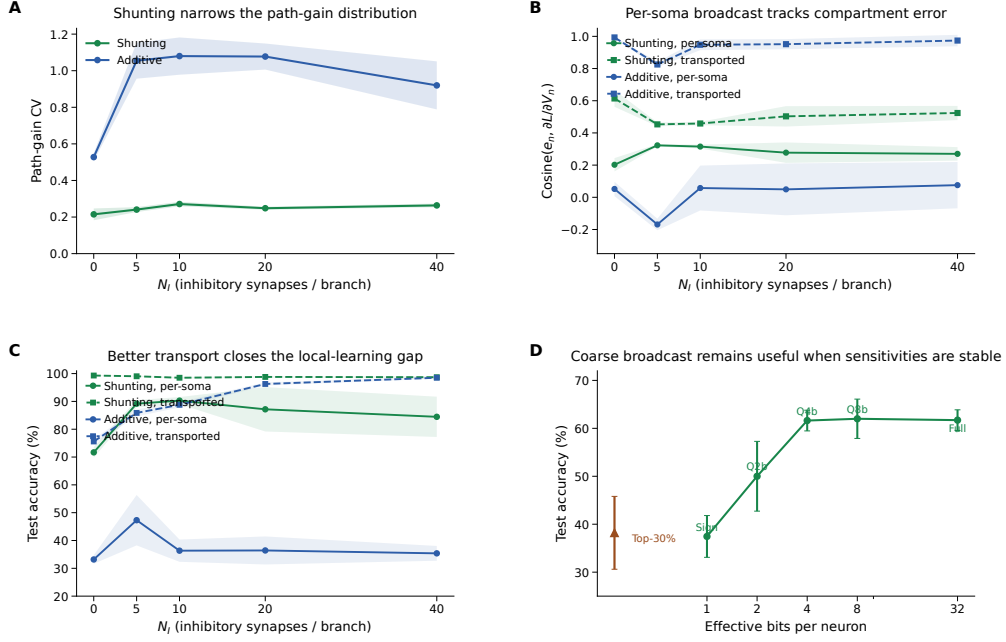


Figure 3: **Mechanistic evidence.** (A) Path-gain dispersion on MNIST, measured as the coefficient of variation of the exact tree attenuation α_n across compartments. Shunting keeps path gains much more concentrated than the additive control as inhibition increases. (B) Direct compartment-error fidelity on noise resilience. Solid lines compare the per-soma broadcast e_n to the exact compartment error $\partial L / \partial V_n$; dashed lines show the transported-error oracle obtained by recursively applying exact path gains. Shunting substantially improves the fidelity of the simple per-soma broadcast, while the transported oracle defines the remaining approximation gap. (C) Learning with the transported-error oracle on noise resilience. Improving the broadcast field is enough to recover near-ceiling local learning, especially in the additive architecture where per-soma broadcast fails most severely. (D) Broadcast bandwidth. Four-bit quantization preserves full performance; only sign-only and sparse modes degrade sharply.

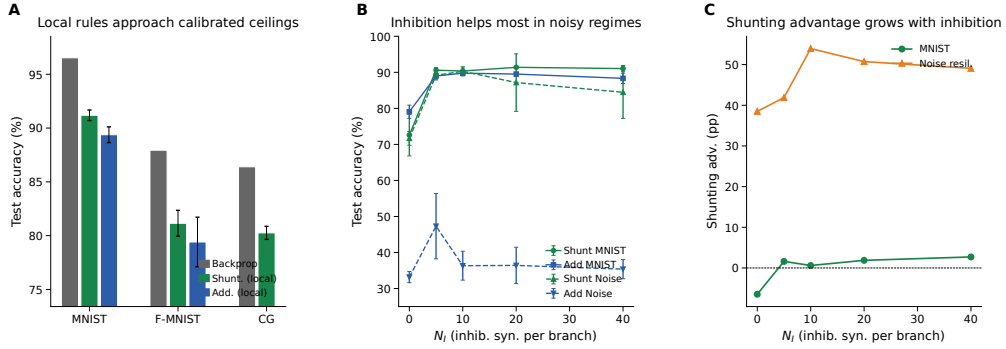


Figure 4: **Local competence and regime dependence.** (A) Backprop ceiling (gray) vs. best local rule (SF per-soma) for shunting (green) and additive (blue) cores on MNIST, Fashion-MNIST, and context gating. (B) N_I dose-response: test accuracy vs. inhibitory synapses per branch (N_I) for shunting and additive cores on MNIST (solid) and noise resilience (dashed). Shunting needs $N_I \geq 5$ to reach high accuracy; additive learning fails on noise resilience regardless of N_I . (C) Shunting advantage (Δ , in percentage points) vs. N_I . The shunting advantage grows with inhibitory synapses per branch and is larger on the noise-resilience task than on MNIST.

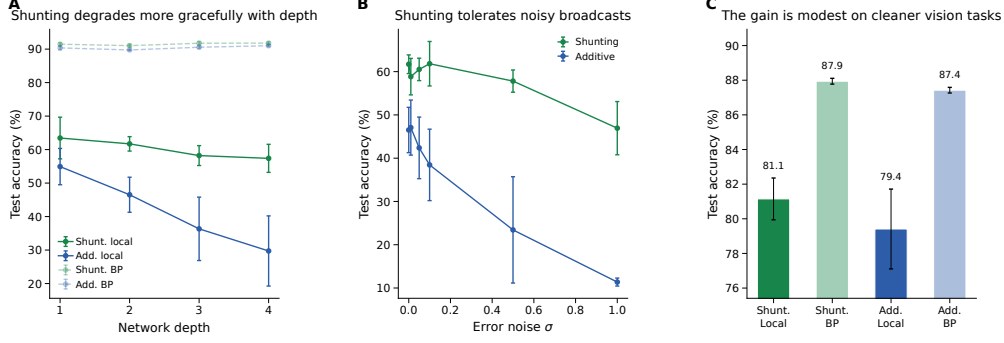


Figure 5: **Additional stress tests.** (A) Depth scaling: shunting local (green) degrades more gracefully from 63.5% to 57.4%; additive local (blue) drops from 54.9% to 29.7%. Backprop ceilings (dashed) remain stable. (B) Broadcast-noise robustness: shunting remains stable for $\sigma \leq 0.1$; additive degrades rapidly and reaches chance at $\sigma = 1$. (C) Fashion-MNIST: the shunting gain is modest on this cleaner benchmark (81.1% local vs. 87.9% backprop), consistent with the regime-dependent story in the main text.

291 4.5 Regime Dependence of the Shunting Advantage

292 The benefit of shunting varies across tasks. On MNIST with matched per-soma broadcast, shunting
 293 outperforms additive by about 2 percentage points; a similar pattern holds on Fashion-MNIST (81.1%
 294 vs. 79.4%). On tasks that require credit signals robust to noise, the gap is much larger: +50.3 pp on
 295 noise resilience (with $N_I=10$ inhibitory synapses per branch) and +24.8 pp on info shunting ($N_I=0$).
 296 Figure 4B–C shows that the advantage grows with inhibitory conductance strength. Additive cores
 297 can still perform reasonably on easier settings, reaching about 89% on MNIST after tuning. However,
 298 they fail in regimes where inhibition-mediated normalization matters for gradient propagation.

299 4.6 Additional Stress Tests

300 We test whether the shunting advantage persists under three stress tests: increased dendritic depth,
 301 noisy broadcast signals, and a second real-world benchmark.

302 **Depth scaling.** Varying dendritic depth from 1 to 4 layers (branch factors [9] to [3, 3, 3, 3]), shunting
 303 local learning degrades from 63.5% to 57.4% while additive local degrades more steeply from 54.9%
 304 to 29.7% (Fig. 5A). Backprop ceilings remain stable at about 90–92% for both architectures. The
 305 shunting advantage grows with depth (+8.5 pp at depth 1 to +27.7 pp at depth 4), consistent with
 306 the path-gain attenuation in Theorem 1: each additional layer introduces an attenuation factor of the
 307 form $R_k^{\text{tot}} g_k^{\text{den}} < 1$. In practice, shunting changes the distribution of these factors and yields better
 308 depth scaling under local learning.

309 **Noise robustness.** Adding Gaussian noise $\mathcal{N}(0, \sigma^2)$ to the broadcast error (Fig. 5B), shunting local
 310 learning is robust to $\sigma \leq 0.1$ (about 62%) and degrades gracefully beyond, while additive drops
 311 from 46.5% to chance at $\sigma=1.0$. This pattern is consistent with the proposed mechanism: shunting
 312 produces better-scaled local sensitivities, so gradient direction is preserved more often even when the
 313 broadcast magnitude is corrupted.

314 **Fashion-MNIST.** On Fashion-MNIST (Fig. 5C), shunting local reaches 81.1% vs. a 87.9% back-
 315 prop ceiling (gap 6.8%); additive local reaches 79.4% vs. 87.4% backprop (gap 8.0%). The shunting
 316 advantage is modest (about 1.7 pp) on this clean classification task, consistent with the regime-
 317 dependent pattern above: the benefit is largest on tasks that require robust credit signals under
 318 noise.

319 4.7 Extension: Structured Feedback When Per-Soma Broadcast Fails

320 The main paper focuses on low-bandwidth scalar or per-soma broadcast. We now consider a harder
 321 cue-integration benchmark where that approximation is insufficient because different dendritic

pathways must be credited differently. The task asks the model to combine two noisy sensory cues using a context signal that indicates which cue is reliable on each trial. In the fixed-pathway control, context is copied into both cue pathways. In the learned-pathway setting, the router must instead discover that separation and align one latent pathway with each cue stream.

Figure 6 summarizes five-seed runs for the hard benchmark together with a focused tuning sweep. Fixed-pathway additive local learning remains near ceiling ($99.1 \pm 0.3\%$), fixed-pathway shunting local learning is strong but more variable ($92.8 \pm 6.1\%$), learned-routing additive local learning also succeeds ($98.1 \pm 2.4\%$), and standard learned-router training stays near $99.2 \pm 0.4\%$ for both additive and shunting cores. The lower accuracy and higher variance for fixed-pathway shunting relative to fixed-pathway additive (92.8 vs. 99.1%) suggests that shunting local learning is already somewhat stressed by pathway duplication: when context is copied into both cue branches, the conductance-dependent sensitivities may introduce ambiguity that scalar broadcast cannot fully resolve. Taken together, these controls show that the benchmark and router are learnable while flagging that shunting LocalCA requires structured feedback even in easier routing conditions.

The specific failure appears when shunting local learning uses the default per-soma broadcast. At the original 90-epoch setting, learned-routing accuracy is only $82.6 \pm 14.2\%$. Lowering router temperature to 0.3 or freezing the encoder for the first 20 epochs changes little ($87.6 \pm 13.3\%$ and $83.8 \pm 13.9\%$), even though the router already becomes highly specialized (mean max assignment $0.98-1.00$). A focused optimization sweep shows that training budget alone is not the issue: extending training to 140 epochs leaves per-soma performance unchanged ($82.5 \pm 14.1\%$), and even lowering the learning rate to 10^{-3} reaches only $88.2 \pm 13.9\%$.

Replacing per-soma broadcast with PV-LocalCA changes the regime entirely. The untuned 90-epoch version already improves to $94.0 \pm 11.0\%$, and the tuned setting (140 epochs, learning rate 10^{-3}) reaches $99.1 \pm 0.3\%$ test and $99.4 \pm 0.2\%$ validation accuracy while maintaining strong specialization (mean max assignment 0.995 ± 0.008). The failure is therefore not caused by pathway discovery itself; it arises because scalar broadcast does not preserve pathway identity once different cue branches must be credited differently. When the feedback preserves that structure, learned-routing shunting local learning becomes reliable.

5 Related Work

High-performance local learning. Several recent methods have substantially raised the benchmark-performance bar for local learning on standard architectures. Auxiliary-network approaches such as AugLocal scale greedy local learning to deep CNNs, including ImageNet-sized models [36]. CCL uses a dual forward/backward network pair to avoid weight transport while maintaining strong CNN and MLP performance [39]. DLL derives local rules from abstract apical/basal compartment structure and reports strong MLP/CNN/RNN results under explicit plausibility criteria [37]. Our aim is therefore not to claim the strongest benchmark accuracy among all local-learning methods, but to identify the biophysical mechanism that determines when low-bandwidth credit signals become useful in conductance-based dendritic networks.

Local credit-assignment dynamics. Random feedback [9], DFA [10], forward-forward [27], PEPITA [28], PAL [35], DTP [29], DFC [30], prospective configuration [17], equilibrium propagation [16], and predictive coding [14, 31] solve locality through diverse feedback or inference dynamics. EBD [38] is especially close in spirit to our work because it asks when direct error broadcast can be made useful by shaping the broadcast field. Our contribution is complementary: rather than changing the inference dynamics, we identify a conductance-based inductive bias—path-gain concentration under shunting—that makes a low-bandwidth broadcast more faithful to the exact compartment error. We do not benchmark these methods directly here because their iterative controllers or inference loops are not architecture-matched to the steady-state conductance cores studied in this paper.

Dendrites for robustness and context. Dendritic trees support nonlinear computation [3, 4, 32] and have inspired biologically plausible learning schemes including segregated dendrites [11], dendritic prediction errors [5, 12], burst-dependent plasticity [24, 25], and latent equilibrium [26]. Outside supervised iid classification, active-dendrite architectures can also improve robustness in dynamic multi-task settings where point-MLP baselines suffer interference [48]. Our cue-routing

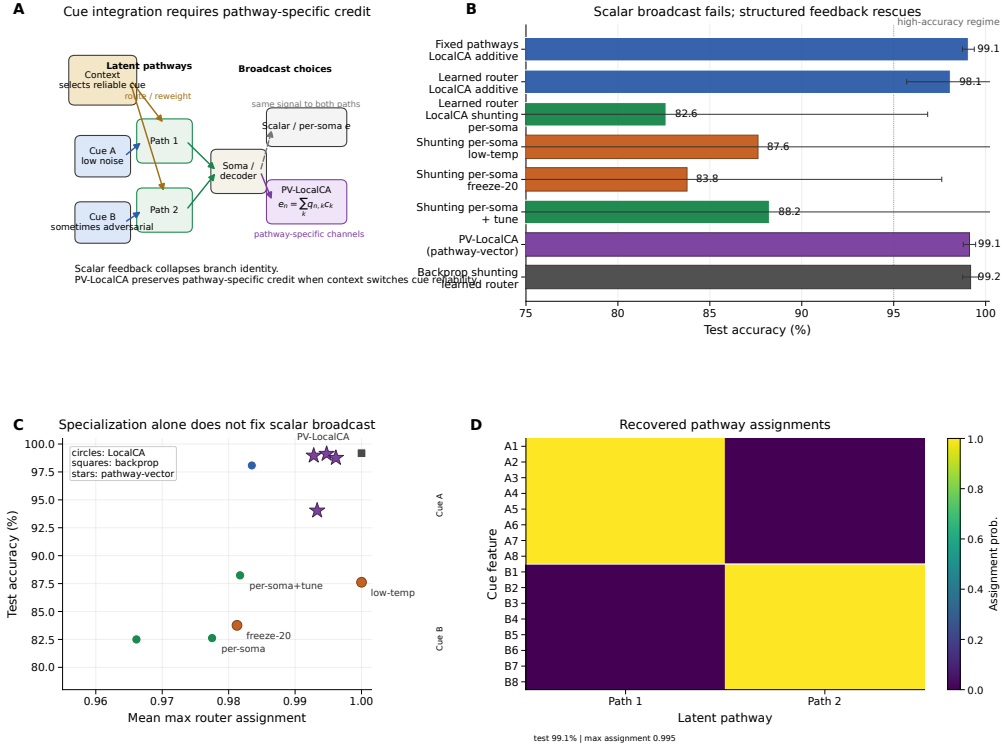


Figure 6: Structured feedback for pathway discovery on cue integration. (A) Task schematic. Context determines which noisy cue is reliable on each trial. Scalar or per-soma broadcast sends the same teaching signal to both pathways, whereas PV-LocalCA preserves pathway identity through a low-rank pathway-vector feedback field. (B) Seed-averaged diagnosis and focused tuning (5 seeds per condition). Learned-routing additive local learning remains high, but learned-routing shunting with per-soma broadcast stays unstable; low-temperature, freeze-20, and per-soma tuning help only modestly, whereas tuned pathway-vector feedback reaches $99.1 \pm 0.3\%$. (C) Test accuracy vs. router specialization across learned-router conditions. High specialization alone does not guarantee success: per-soma, low-temperature, and freeze-20 runs remain inaccurate despite near-discrete assignments. (D) Recovered pathway assignments for a representative tuned pathway-vector run. Cue-A and cue-B features segregate cleanly into separate latent pathways.

extension speaks to the same broader question—when dendritic structure helps preserve pathway-specific computation—but uses conductance-based normalization and structured credit signals rather than sparse kWTA gating.

Neuroscience grounding. Shunting implements divisive normalization [6], though its rate effects can be subtractive in some regimes [7]; inhibitory conductance also modulates gain and signal-to-noise [33]. Beniaguev et al. [34] showed that single cortical neurons can support deep-like computations, motivating dendrites as learning substrates rather than mere architectural decoration. Recent experimental evidence for vectorized instructive signals in cortical dendrites [47] further supports our structured-feedback extension: when different branches should be credited differently, biology may already provide more than a single scalar somatic teaching signal. To our knowledge, prior work has not directly tied gradient-fidelity diagnostics, exact path gains, and shunting inhibition into one causal account of when local dendritic learning works.

6 Discussion

We started from exact gradients for a conductance-based dendritic tree and asked when a somatic broadcast can serve as a useful proxy for compartment-specific errors. In the models studied here, the answer depends strongly on the inhibitory regime. Shunting changes the conductance denominator

and stabilizes effective local sensitivities, improves agreement between local and backprop gradients, and leads to better local learning than an additive control, especially in noisy or inhibition-dependent settings. Explicit normalization recovers part of this benefit but not all of it, suggesting that shunting changes more than just overall scale.

Limitations. Our best accuracy (91.4% MNIST, 81.1% Fashion-MNIST) is below several recent local-learning methods that operate in abstract compartments or standard activation spaces [36–39]. This reflects the constraints of operating in conductance-based voltage space: bounded voltages, positive conductances, and a denominator-heavy computation graph. In the conservative flattened-CIFAR-10 sweep shown in Appendix Fig. S6, shunting local learning reaches 21.2% vs. 40.1% backprop, indicating that the gradient-fidelity mechanism alone is insufficient for harder tasks without architectural advances (e.g., convolutional encoders). That trade-off reflects our emphasis on mechanism rather than absolute benchmark performance: the gradient-fidelity analysis helps explain why some architectures support local learning better than others. Second, naive local broadcasts remain much weaker than per-soma on MNIST, and the hard cue-routing benchmark requires an even more structured pathway-vector signal (Sec. 4.7); recent experimental findings on vectorized instructive signals in cortical dendrites [47] support the biological plausibility of such structured feedback. Our claims are therefore specific to the broadcast family studied here rather than arbitrary local mismatch heuristics. Per-soma broadcast should be understood as low-bandwidth but still neuron-global: it assumes access to the full somatic error vector and therefore sets the current locality boundary of the method. Third, scaling to deeper architectures degrades local learning more than backprop (Fig. 5A; Appendix J), indicating that depth-dependent credit attenuation remains challenging. At depth 4, shunting local learning is still roughly 33 percentage points below backprop, so the present rules improve deep credit assignment but do not solve it. Fourth, shunting networks are *not* gain-invariant in practice: exploratory gain-shift pilots show that point MLPs (scale-equivariant via ReLU) generalize nearly perfectly to large global input gains, while shunting networks degrade sharply because the leak conductance breaks exact gain invariance. We therefore treat gain robustness as distinct from the credit-assignment mechanism emphasized in the main paper. Fifth, standard FA is incompatible with dendritic layer geometry (Appendix Fig. S5), limiting the space of applicable feedback methods. We also do not directly compare against iterative target-propagation or equilibrium-style methods, because a clean comparison would require re-implementing their inference dynamics inside the same conductance-based dendritic architecture rather than benchmarking against separate abstract models. Finally, the weight-statistics analysis is secondary to the main credit-assignment claim and is reported in Appendix G; it should be read as descriptive consistency rather than as core evidence for the learning mechanism.

Broader relevance. More broadly, the results are relevant to neuroscience, machine learning, and neuromorphic hardware. For neuroscience, they suggest a connection between divisive normalization and local credit fidelity, extending its known roles in gain control [33] and sensory coding [6]. For machine learning, they show that conductance-based inductive biases can shape gradient geometry in ways that benefit local learning. For neuromorphic implementations, the update rules remain strictly local and parallelizable.

Potential harmful consequences. The work is primarily methodological and does not introduce a deployable decision system or a new human dataset. Still, more efficient local learning rules for adaptive hardware could eventually be incorporated into surveillance, autonomous, or other safety-critical systems. We therefore view the most plausible risks as downstream misuse of efficient on-device learning rather than harm from the present experimental setup itself.

Testable predictions. Our framework predicts that (1) stronger perisomatic inhibition yields more precise synaptic plasticity; (2) GABA_A blockade selectively impairs multi-layer credit tasks; (3) the dendritic–somatic voltage correlation (ρ_n) increases during learning; (4) learned excitatory weight distributions follow log-normal statistics whose width depends systematically on inhibitory conductance (Appendix G). These predictions can help distinguish our account from models where dendrites mainly serve as computational substrates.

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Dataset	Rule	Top-10 valid	Top-10 test
MNIST	3F	0.611	0.622
MNIST	4F	0.620	0.628
MNIST	5F	0.912	0.916
Context gating	3F	0.398	0.396
Context gating	4F	0.411	0.411
Context gating	5F	0.807	0.789

Table S1: **Rule-family ranking.** Top-10 mean across completed local-competence sweeps.

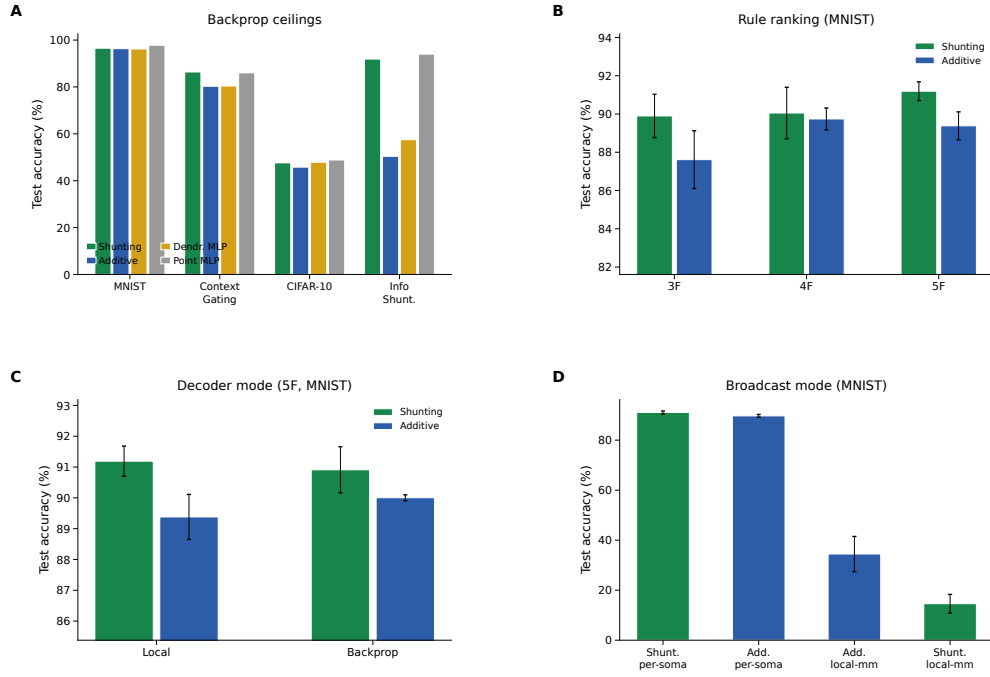


Figure S1: **Capacity calibration (supplementary).** (A) Phase 1 backprop ceilings across all architectures and datasets. (B) Rule-family ranking: 5F consistently outperforms 4F and 3F. (C) Decoder mode comparison (local vs. backprop decoder, 5F MNIST). (D) Broadcast mode comparison: per-soma is required for strong performance; local-mismatch fails.

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Component	Representative effect	Interpretation
Shunting architecture	Path-gain CV stays at about 0.21–0.27 under shunting vs. 0.53–1.08 under additive; on noise resilience, per-soma local learning rises from roughly 0.33–0.47 (additive) to 0.72–0.90 (shunting).	This is the main architectural effect in the paper: shunting concentrates path gains and yields a more faithful low-bandwidth credit signal.
Broadcast quality	On MNIST (5F), shunting per-soma reaches 0.912 ± 0.005 , while local-mismatch remains at 0.146 ± 0.046 .	Correctly matching the broadcast field to the somatic error matters far more than arbitrary local mismatch heuristics.
Transported oracle	On noise resilience, additive local learning rises from 33–47% under per-soma to 75.6–98.5% under transported broadcast; shunting rises from 71.7–90.3% to 98.5–99.3%.	The oracle is the strongest causal evidence that broadcast fidelity, not just architecture, is the dominant bottleneck.
Rule hierarchy	In the competence sweeps, MNIST improves from 0.622 (3F) and 0.628 (4F) to 0.916 (5F); context gating improves from 0.396/0.411 to 0.789.	5F is the strongest practical rule in this paper, though its extra factor remains heuristic rather than theorem-derived.
5F stability	Tightening the ϕ clamp to $[0.5, 2.0]$ changes MNIST by only about +0.4 pp; varying EMA from 0.05 to 0.20 changes it by less than 0.2 pp.	The practical gain from 5F is not especially brittle to its main tuning knobs.
Structured feedback	On learned cue routing, tuned shunting per-soma reaches $88.2 \pm 13.9\%$, while tuned pathway-vector feedback reaches $99.1 \pm 0.3\%$.	Structured feedback matters when different branches must be credited differently; it is not a generic replacement for per-soma broadcast on ordinary tasks.

Table S2: **Which components carry the observed gains?** A compact summary of the empirical division of labor across architecture, broadcast design, and LocalCA rule variants.

Core	Broadcast	Decoder	Test (mean $\pm$ std)
Shunting	per-soma	local	0.912 $\pm$ 0.005
Shunting	per-soma	backprop	0.909 $\pm$ 0.008
Shunting	local-mismatch	local	0.146 $\pm$ 0.046
Shunting	local-mismatch	backprop	0.146 $\pm$ 0.037
Additive	per-soma	local	0.894 $\pm$ 0.007
Additive	per-soma	backprop	0.900 $\pm$ 0.001
Additive	local-mismatch	local	0.342 $\pm$ 0.058
Additive	local-mismatch	backprop	0.348 $\pm$ 0.095

Table S3: **Local-mismatch recheck (MNIST, 5F)**. Per-soma is consistently strong; local-mismatch remains much weaker.

Condition	Metric	Value
Decoder: local vs backprop vs frozen (MNIST, sandbox)	test acc	0.379 vs 0.379 vs 0.176
Shunting vs additive, matched (MNIST, sandbox)	Δ test	+0.210
Per-soma, path on vs off (synthetic task, sandbox)	test / MI(E,I;C)	-0.003 / +0.017

Table S4: **Mechanistic ablations**. Results from controlled small-network sandbox experiments using minimal architectures to isolate individual factors. Sandbox runs use a small 2-layer dendritic network (branch factor [9]) trained for 30 epochs; they are not directly comparable to the headline capacity-calibrated results.

A Supplementary Results

Rule-Family Ranking

Which Components Carry?

Broadcast Mode Comparison and Local-Mismatch Recheck

Phase 1 Capacity Ceilings

Ablation Results

Extended Gradient Analysis and N_I Sweep Detail

Verification and Reproducibility

FA/DFA Baseline Comparison

CIFAR-10 Results

Additive Normalization Control

B Implementation Details

Biological Plausibility Assumptions

Each synapse has access to: presynaptic activity x_j (local), compartment voltage V_n (local membrane potential), reversal potential E_j (fixed by receptor type), and total input resistance $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$ (computable from local conductances). The 4F modulator ρ_n requires online estimation of voltage covariance (biologically plausible via slow calcium signals); the 5F factor ϕ_n requires a linear regression proxy (implementable via eligibility traces). The only non-local quantity is the broadcast error e_n , which requires a top-down or neuromodulatory signal from the soma to dendritic compartments. We assume per-neuron resolution (one error per output neuron) for the per-soma broadcast mode.

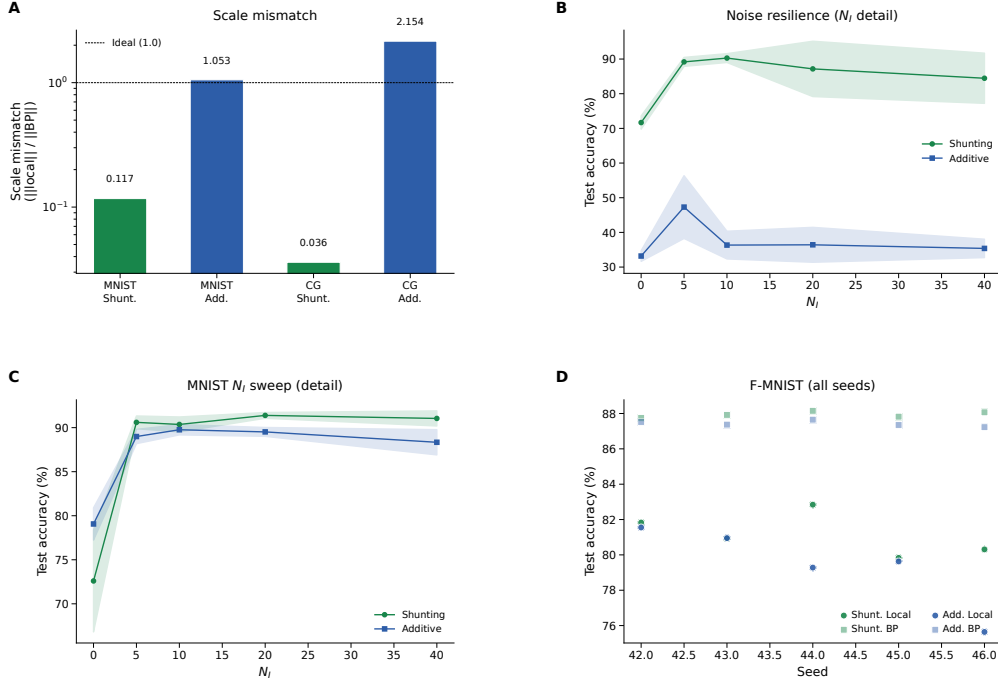


Figure S2: **Extended gradient and N_I analysis (supplementary).** (A) Scale mismatch bars: shunting achieves near-ideal scale (0.117); additive exhibits order-of-magnitude distortion (> 1.0). (B) Noise resilience N_I dose-response with error bands (± 1 s.d.). (C) MNIST N_I dose-response detail with error bands. (D) Fashion-MNIST individual seeds for all conditions, showing consistency across runs.

Quantity	Symbol	Convention
Voltage	V	Normalized to $[-1, 1]$
Conductances	$g^{\text{syn}}, g^{\text{den}}$	Nonneg. via softplus
Leak conductance	g^{leak}	Set to 1
Input resistance	R^{tot}	≤ 1

Table S5: Units and normalization.

555 Units and Parameterization

556 Decoder Update Modes

557 W_{dec} maps $V_L \rightarrow \hat{y}$. Modes: **backprop** ($\nabla_W L$ via autograd), **local** ($\Delta W = \eta \langle \delta_0 V_L^\top \rangle_B$), **frozen**
558 ($\Delta W = 0$).

559 Model and Mode Inventory

560 LocalCA Extension Compatibility

561 Compute Resources

562 All experiments were run on an internal GPU cluster. Most MNIST, Fashion-MNIST, and synthetic-
563 task runs used a single NVIDIA A100 or H100 GPU and finished in minutes to tens of minutes per
564 seed; the flattened CIFAR-10 runs required longer single-GPU jobs but still did not use distributed or
565 multi-GPU training. Reported headline metrics are aggregated over 3–5 seeds as stated in the text;
566 the broader exploratory sweeps used to tune and diagnose the method required additional compute
567 beyond the final reported runs.

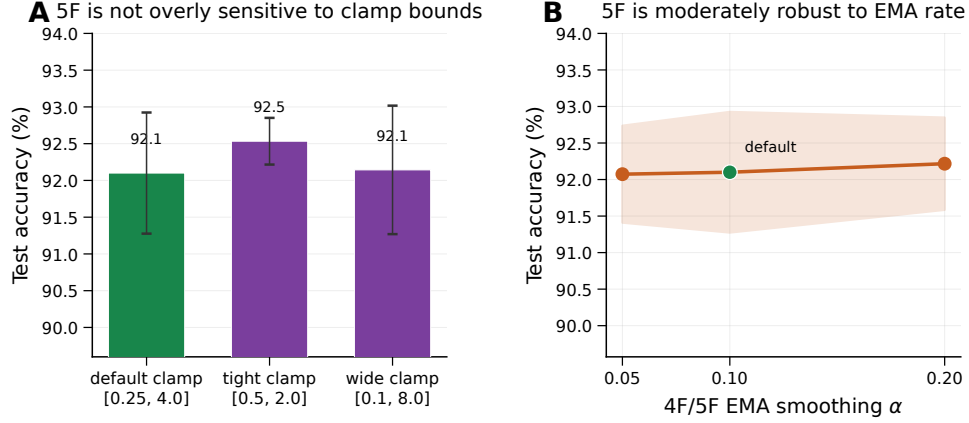


Figure S3: **5F sensitivity analysis (MNIST, shunting, per-soma broadcast)**. **(A)** Sensitivity to the ϕ_n clamp interval. The default clamp [0.25, 4.0] is already strong ($92.1 \pm 0.8\%$); tightening it to [0.5, 2.0] slightly improves performance ($92.5 \pm 0.3\%$), while widening it to [0.1, 8.0] is effectively neutral ($92.1 \pm 0.9\%$). **(B)** Sensitivity to the 4F/5F EMA rate used for online statistics. Varying α from 0.05 to 0.20 changes mean test accuracy by less than 0.2 pp. Together these results suggest that the 5F gain is not driven by a brittle single hyperparameter choice, even though the factor itself remains heuristic. Errors: ± 1 s.d. across 3 seeds.

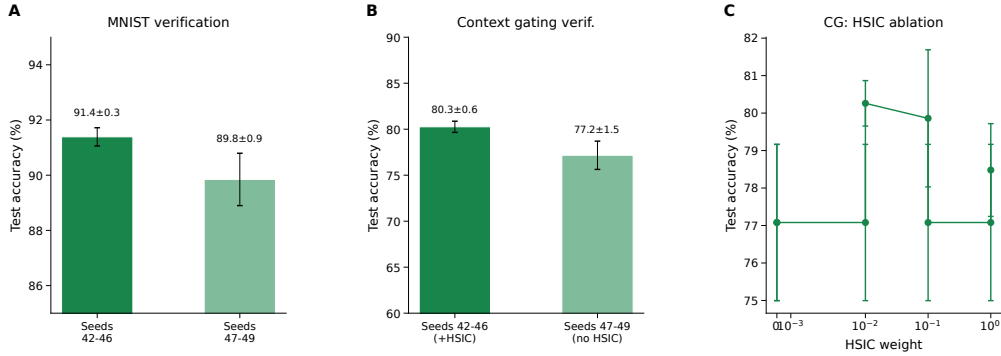


Figure S4: **Verification and reproducibility (supplementary)**. **(A)** MNIST verification: main seeds (42–46) yield $91.4 \pm 0.3\%$; held-out seeds (47–49) yield $89.8 \pm 0.9\%$, confirming generalization. **(B)** Context gating verification: main seeds (+HSIC) $80.3 \pm 0.6\%$; held-out seeds (no HSIC) $77.2 \pm 1.5\%$. The roughly 3 pp gap reflects HSIC removal, not seed sensitivity. **(C)** HSIC weight ablation on context gating: moderate weights (0.01–0.1) perform best.

568 Code, Data, and Asset Availability

569 MNIST, Fashion-MNIST, CIFAR-10, and MICrONS are public datasets or public benchmark assets
 570 cited in the paper; we use them under their standard published access conditions and cite their original
 571 sources in the bibliography. The present anonymous submission does not include a public code URL
 572 in order to preserve double-blind review, but the paper specifies the model family, training modes, key
 573 hyperparameters, and representative configurations, and we plan to release code and configuration
 574 files after review. The paper does not release a new dataset or a stand-alone pretrained model asset.

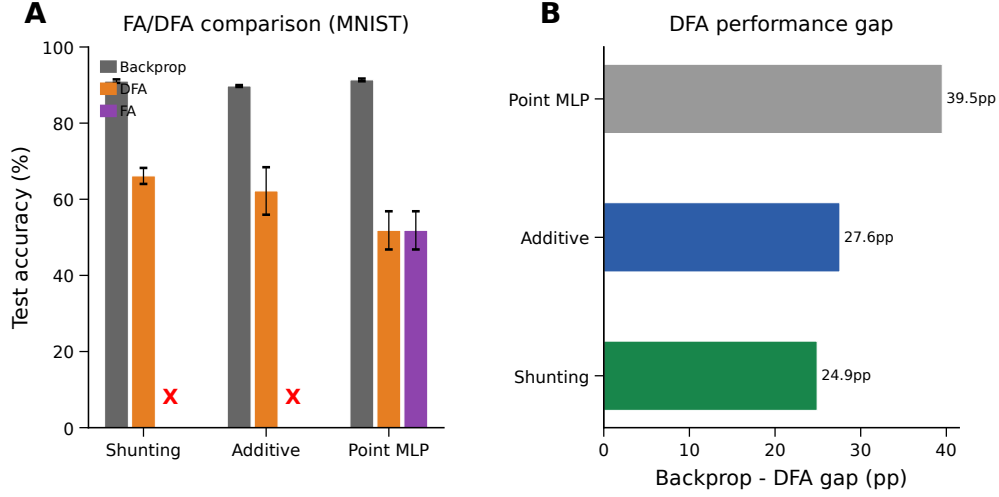


Figure S5: **Feedback alignment baselines (MNIST).** (A) Grouped comparison: standard backprop (gray), DFA (orange), and FA (purple). Red X marks indicate FA failure on dendritic architectures (dimensional incompatibility between random feedback matrices and block-structured dendritic layers). DFA achieves 66.1% on shunting vs. 62.2% additive vs. 51.8% point MLP. (B) Backprop–DFA gap: dendritic architectures (24.9–27.6 pp) show smaller gaps than point MLPs (39.5 pp), suggesting conductance-based architecture is partially compatible with random feedback.

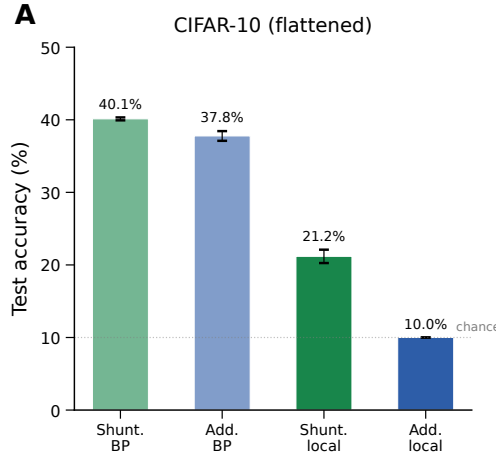


Figure S6: **Conservative flattened CIFAR-10 reference sweep.** In this appendix stress test, shunting backprop reaches 40.1%, additive backprop 37.8%, shunting local 21.2%, and additive local 10.0% (chance). The backprop–local gap (about 19 pp for shunting) is larger than on MNIST, reflecting the increased difficulty of propagating credit through conductance-based layers for complex visual features. This panel should be read as a conservative flattened-image benchmark rather than as the tuned CIFAR ceiling for the architecture family.

575 Algorithm

Algorithm 1 Local Credit Assignment

- 1: **Input:** Model, batch (x, y) , config $\mathcal{C}$
- 2: Forward pass; loss L , output error δ^y
- 3: Somatic error $\delta_0 = W_{\text{dec}}^\top \delta^y$
- 4: **for** each layer n (reverse) **do**
- 5: $e_n = \text{broadcast}(\delta_0, \mathcal{C})$; optionally gate by pathway roles
- 6: Compute ρ_n, ϕ_n (EMA estimators, if enabled)
- 7: Apply the configured local update (Eq. 6 or 7)
- 8: **end for**
- 9: Clip gradients; optimizer step

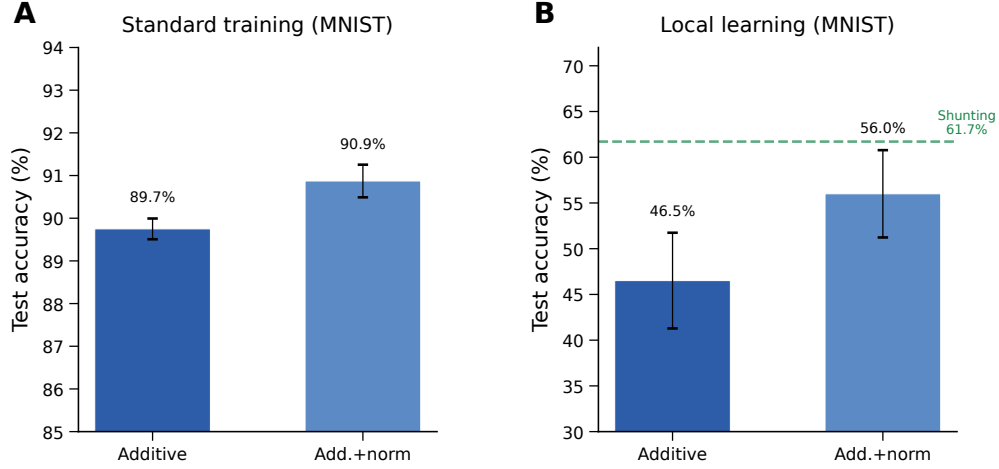


Figure S7: **Additive + normalization control (MNIST)**. (A) Standard backprop: normalization provides a small boost (89.7% $\rightarrow$ 90.9%). (B) Local learning: normalization improves additive from 46.5% to 56.0% (+9.5 pp), partially closing the gap to shunting (61.7%, dashed green). The remaining gap (−5.7 pp) indicates shunting provides benefits beyond simple voltage normalization, including bounded activations, conductance-dependent input weighting, and biophysically constrained sensitivity structure.

Category	Implemented modes	Role in this paper
Core models	point MLP; dendritic MLP; dendritic additive; dendritic shunting; pathway-routed dendritic cores	Main comparisons use dendritic additive vs. dendritic shunting; routed cores are used for cue integration.
Training strategies	standard backprop; LocalCA; DFA; FA; transported-broadcast oracle	Main text uses standard and LocalCA; DFA/FA remain appendix baselines; path transport is an oracle upper bound.
Broadcast modes	scalar; per-soma; local mismatch; low-rank; path transport; pathway vector	Main text centers on per-soma and pathway-vector; scalar and local-mismatch are supporting controls; path transport is diagnostic only.
Recent dendritic extensions	path propagation; router-derived branch roles; pathway-vector feedback; inhibitory homeostasis	These were added to test morphology-aware credit transport and structured branch-specific plasticity beyond the original 3F/4F/5F family.

Table S6: Implemented model families and LocalCA operating modes. Only a subset is used for the main claims; the remaining modes serve as controls, ablations, or structured extensions.

C Theoretical Details

Variant Taxonomy

Biological Analogs

D Morphology-Aware Extensions

Path-integrated propagation. Modulate broadcast error by $\pi_n = \pi_{n-1} \cdot R_{n-1}^{\text{tot}} \cdot \bar{g}_{n-1}^{\text{den}}$, approximating depth attenuation from Eq. 4.

Depth modulation. Per-branch scaling $\rho_j = \rho_{\text{base}} / (d_j + \alpha)$, mirroring cable attenuation.

Dendritic normalization. $\Delta g_j^{\text{den}} \leftarrow \Delta g_j^{\text{den}} / (\sum_k g_k^{\text{den}} + \varepsilon)$, analogous to homeostatic scaling [23].

Extension	Meaning	Compatible with	Typical use here
Per-soma broadcast	Somatic error vector shared across compartments	3F, 4F, 5F	Default main-text LocalCA setting
Low-rank broadcast	Small number of broadcast channels instead of a scalar or full field	3F, 4F, 5F	Intermediate control when rank-1 is too restrictive
Path transport	Recursive tree-transported error using path gains	3F, 4F, 5F	Oracle upper-bound broadcast
Path propagation	Approximate path-gain attenuation applied to broadcast	3F, 4F, 5F	Morphology-aware modifier for deeper trees
Pathway vector	Structured branch/pathway-specific broadcast field	3F, 4F, 5F	Used mainly in PV-LocalCA on routed tasks
Branch-role rules	Scales updates by router- or pathway-derived branch specialization	3F, 4F, 5F	Used in the structured pathway extension
Inhibitory homeostasis	Auxiliary local inhibitory balancing term	3F, 4F, 5F	Optional stabilizer in routed or shunting settings

Table S7: Compatibility of the newer LocalCA extensions with the base 3F/4F/5F family. These extensions mostly modify the error field or branch scaling rather than defining a new “6F” or “7F” rule.

Rule	Factors	Cost	Best regime
3F	$x, (E - V), e$	$\mathcal{O}(1)$	Baseline
4F	$3F + \rho$	$\mathcal{O}(1)$	Improved conditioning
5F	$4F + \phi$	$\mathcal{O}(d_n)$	Strongest overall
5F+VH	$5F + \text{local voltage homeostasis}$	$\mathcal{O}(d_n)$	Stabilized LocalCA / warm-up
PV-LocalCA	$3F + \text{structured } e_n + \text{role alignment}$	$\mathcal{O}(r_n)$	Pathway discovery

Table S8: Variant taxonomy. ‘5F+VH’ denotes the same base 5F rule with an additional strictly local voltage-centering auxiliary that can also be scheduled through multi-stage LocalCA warm-up; the codebase also supports this as a shorthand ‘5f_vh’ alias in configuration files.

Pathway-vector broadcast. For tasks with latent pathway structure, the broadcast becomes a low-dimensional vector over router-inferred pathway roles. This vector is gated by local pathway activity and propagated through the block-structured tree so that earlier branches receive feedback aligned with their downstream descendants.

Router-derived branch roles. Each branch receives a role profile inferred from router assignments and incoming block weights rather than from a hand-coded apical/basal label. Plasticity is then modulated by branch selectivity and a synapse-specific role-alignment factor, emphasizing pathway-consistent updates without imposing a heuristic branch taxonomy.

E HSIC Auxiliary Objectives

Following [18], we use kernel-based HSIC objectives. Self-decorrelation: $\mathcal{L}^{\text{self}} = B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Z \mathbf{H})$. Target-correlation: $\mathcal{L}^{\text{target}} = -B^{-2} \text{tr}(\mathbf{K}_Z \mathbf{H} \mathbf{K}_Y \mathbf{H})$. Moderate weights (0.01–0.1) help on context gating; negligible on MNIST. Online statistics (ρ_n, ϕ_n) use Welford’s algorithm [22].

F Online Variant with Eligibility Traces

Continuous-time eligibility: $\tau_e \dot{e}_j^{\text{syn}} = -e_j^{\text{syn}} + x_j(E_j - V_n)R_n^{\text{tot}}$. Update: $\Delta g_j^{\text{syn}} \propto \int e_j^{\text{syn}}(t)m_n(t) dt$ [19, 20].

Component	Analog	Interpretation
R_n^{tot}	Input resistance	Sensitivity modulation
$(E_j - V_n)$	Synaptic driving force	Local gradient factor
Shunting	Divisive normalization	$\partial V / \partial g_I \propto -V$
ρ_n	Layer relevance	Output correlation
ϕ_n	Signal propagation	Conditional predictability
Pathway vector	Branch-specific top-down drive	Structured teaching signal
Role alignment	Pathway-selective plasticity	Branch specialization

Table S9: Biological analogs.

Condition	MNIST		Fashion-MNIST	
	Exc σ	Inh σ	Exc σ	Inh σ
BP + Shunting	0.94±0.01	0.87±0.01	0.92±0.01	0.84±0.01
BP + Additive	1.30±0.01	1.22±0.01	1.16±0.01	1.05±0.01
Local + Shunting	1.22±0.01	1.71±0.02	1.13±0.06	1.36±0.06
Local + Additive	1.69±0.16	1.57±0.07	1.43±0.16	1.11±0.12
Song et al. 2005	$\sigma = 0.94$ (rat V1 L5, EPSP amplitude)			
MICrONS E→E	$\sigma = 1.14$ (mouse V1, synapse size, $n=3.9\text{M}$)			
MICrONS I→E	$\sigma = 0.92$ (mouse V1, synapse size, $n=3.5\text{M}$)			

Table S10: **Weight distribution width.** Log-normal σ of excitatory and inhibitory synaptic weights (mean±std across 5 seeds). Biological references are included only as descriptive anchors. With biologically constrained E/I ratios (4:1), shunting self-normalization produces weight heterogeneity in the range spanned by those references, whereas additive integration is broader and less stable.

G Weight Statistics and Biological References

As a secondary analysis, we examine whether the conductance-based update naturally induces characteristic weight statistics. Because synaptic updates scale with input resistance $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$, the voltage equation implies multiplicative weight dynamics, creating a regime analogous to exponentiated gradient descent [43]. Such multiplicative dynamics suggest log-normal weight distributions in trained networks [41, 42]. We compare the resulting distributions against two biological references: electrophysiological data from Song et al. [41] (rat V1 L5 pyramidal, $\sigma = 0.94$) and the MICrONS mm³ connectome [44] (mouse V1 synapse sizes).

Model weight distributions are log-normal. We fit log-normal distributions to the active (top- K masked) excitatory synaptic weights after training on MNIST and Fashion-MNIST. Table S10 shows the fitted σ (standard deviation of $\log w$) for each condition. Shunting consistently produces narrower distributions than additive integration, and backpropagation produces narrower distributions than local learning. These two factors combine approximately additively: BP+Shunting → narrowest, Local+Additive → broadest.

Shunting is closest to the biological references considered here. BP+Shunting produces $\sigma = 0.94$ on MNIST, very close to the Song et al. reference ($\Delta = 0.008$). Local+Shunting produces $\sigma = 1.13$ – 1.22 , close to MICrONS E→E synapse sizes ($\sigma = 1.14$; $\Delta \leq 0.08$). By contrast, Local+Additive produces $\sigma = 1.43$ – 1.69 , broader than any biological reference considered. This pattern is consistent across datasets (MNIST and Fashion-MNIST), suggesting it reflects structural properties of the learning rule rather than task-specific features.

The match depends on inhibitory conductance. We test sensitivity to the E/I synapse ratio using a sweep over $N_I \in \{0, 2, 5, 10, 20, 30, 40\}$ inhibitory synapses per branch (Appendix Fig. S8). Without inhibition ($N_I=0$), all conditions produce $\sigma > 2.0$, far from biology. With biologically constrained ratios ($N_I=10$ – 20 , corresponding to E/I ratios of 4:1 to 2:1; [46]), shunting networks enter the biological range ($\sigma \approx 0.9$ – 1.2), while the effect of N_I on additive networks is less systematic. The weight distribution shape is also robust to dendritic depth (branch factors [9] through [3,3,3,3];

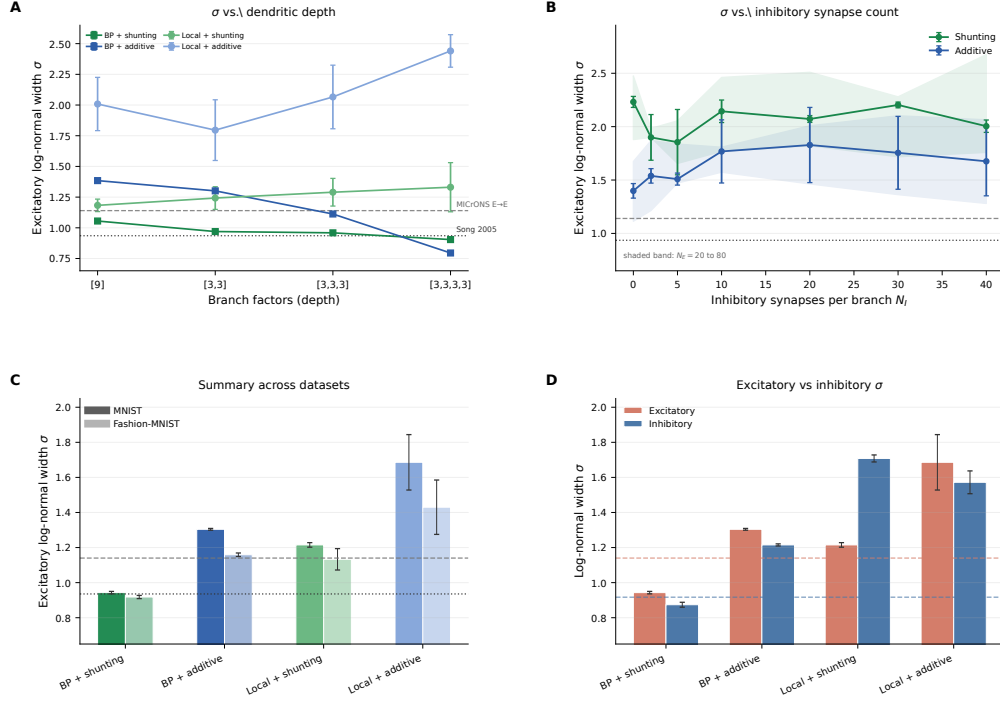


Figure S8: Synaptic weight distribution sensitivity to architecture. (A) Log-normal σ vs. dendritic depth (branch factors [9] to [3,3,3,3]). Both backprop and local shunting remain closer to the biological-width references than their additive counterparts across depth. (B) σ vs. inhibitory synapses per branch (N_I). Without inhibition ($N_I=0$), all conditions are overly broad; with biologically constrained $N_I=10$ –20, shunting enters the biological range. Shaded bands show the sweep over $N_E=20$ to 80. (C) Summary of MNIST and Fashion-MNIST results with biological references. The shunting ordering is consistent across datasets. (D) Excitatory vs. inhibitory weight σ on MNIST with MICrONS E→E and I→E reference bands.

see Appendix Fig. S8A): BP+Shunting converges toward Song et al. with increasing depth, while Local+Shunting remains near MICrONS E→E across depths.

H Weight Distribution Sensitivity Analysis

MICrONS connectome analysis. We extracted synapse size distributions from the MICrONS mm³ condensed connectome (v1181; 9.0M connections, 71.9K neurons; 89% excitatory). Splitting by pre/post cell type: E→E connections ($n=3.85$ M) show $\sigma_{\text{size}}=1.14$ and CV=1.01; I→E ($n=3.53$ M) show $\sigma=0.92$ and CV=1.03; E→I ($n=1.11$ M) show $\sigma=0.83$; I→I ($n=0.51$ M) show $\sigma=0.85$. Inhibitory connections are more frequently multi-synaptic (I→E: 37.5% multi-synapse; E→E: 11.5%), and inhibitory synapses are on average smaller (median 3532 vs. 4676 for E→E).

Sensitivity to ee_synapses. Using the E/I grid sweep ($N_E \in \{20, 40, 80\} \times N_I \in \{0, 2, 5, 10, 20, 30, 40\}$; 168 models), excitatory σ increases modestly with N_E in both architectures (+0.4–0.6 from $N_E=20$ to 80), but the shunting–additive ordering is preserved at all N_E values. The dominant driver of σ is N_I : going from $N_I=0$ to $N_I=10$ reduces shunting σ by approximately 1.0–1.5 units, while the effect on additive is smaller and less systematic.

Sensitivity to depth. From the depth-scaling sweep ([9], [3,3], [3,3,3], [3,3,3,3]; 80 models, 5 seeds each), BP+Shunting excitatory σ decreases with depth (1.06 → 0.90), converging toward Song et al. (0.94). Local+Shunting σ increases slightly (1.18 → 1.33) but remains in the biological range at shallow depths. BP+Additive shows the strongest depth sensitivity (1.38 → 0.79), while Local+Additive diverges with depth (2.01 → 2.44). The morphology-aware modulators (path propagation, depth/centrality weighting) have negligible effect on σ .

Mechanistic interpretation. The shunting conductance $g_n^{\text{tot}} = g_n^{\text{leak}} + \sum g^{\text{exc}} + \sum g^{\text{inh}}$ appears in the update factor $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$. As inhibitory conductance increases, g_n^{tot} grows and R_n^{tot} shrinks, compressing the multiplicative scale of weight updates. This is analogous to a learning rate that self-adjusts via the total conductance: strong synapses contribute more to g_n^{tot} , reducing their own future update magnitude. This acts as a natural form of weight-dependent regularization. The resulting equilibrium distribution is log-normal, consistent with Loewenstein et al. [45] (multiplicative spine dynamics) and Saxena et al. [43] (exponentiated gradient). Additive integration lacks this self-normalizing mechanism, producing broader distributions that fall outside the biological range.

I Synthetic Benchmark Descriptions

Each synthetic task is generated procedurally from a fixed random seed and uses fixed train/val/test splits for reproducibility. All tasks use 500-dimensional flattened inputs and 10-class outputs unless otherwise noted.

Context gating. Inputs are Gaussian mixtures over 10 classes, but the class-conditional means are linearly transformed by one of $C = 4$ context configurations. A one-hot context vector (appended to the input) indicates which configuration is active. Without the context, classes are ambiguous; with it, the optimal boundary shifts. This task tests whether credit assignment can exploit the context gating structure to modulate branch-specific representations appropriately. The HSIC auxiliary objective (weight 0.01) is used to decorrelate layer representations, adding roughly 3 pp under local learning.

Noise resilience. Inputs are MNIST features corrupted by structured channel noise: a random 50×50 projection matrix adds correlated noise drawn from $\mathcal{N}(0, \sigma_{\text{task}}^2 I)$ with $\sigma_{\text{task}} = 1.5$. The noise projection is fixed across training; the network must filter structured interference to recover the underlying class signal. Credit assignment must remain coherent under the corrupted signal, making this task particularly sensitive to path-gain dispersion.

Info shunting. Inputs contain two additive streams: a low-frequency signal and a high-frequency distractor drawn with equal energy. A second context input (separate channel) indicates whether the relevant signal is low- or high-frequency on each trial. Correct classification requires the network to gate the distractor stream, a computation that naturally invites inhibitory modulation. Performance at $N_I=0$ is notably below shunting performance (+24.8 pp), reflecting the task’s dependence on divisive normalization.

Cue integration. Two noisy cue streams (A and B) are presented simultaneously, each carrying partial information about one of 10 classes. A binary context signal indicates which cue is reliable on each trial (cue A or cue B). The fixed-pathway variant duplicates the context into both cue branches; the learned-routing variant requires the router to discover the cue separation from data. This task is used in Sec. 4.7 to test structured pathway-vector broadcast.

J Depth Scaling and Noise Robustness

Depth scaling. Varying dendritic depth from 1–4 layers (branch factors [9] to [3, 3, 3, 3]): shunting local degrades from 63.5% to 57.4%; additive local degrades more steeply from 54.9% to 29.7%. The shunting advantage grows with depth (+8.5 $\rightarrow$ +27.7 pp). Backprop ceilings remain stable at about 90–92%. See Fig. 5A.

Noise robustness. Gaussian noise $\mathcal{N}(0, \sigma^2)$ on broadcast error: shunting is robust to $\sigma \leq 0.1$ (about 62%) while additive drops from 46.5% to chance at $\sigma=1.0$, confirming that shunting credit signals carry useful learning information beyond the broadcast magnitude. See Fig. 5B.

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